CPSC 476 Edge AI – Principles and Practices
Department of Computer Science
California State University, Fullerton
Ning Chen, Ph.D., Professor
Fall 2026

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Office Hours:
CS-546, Monday and Wednesday 11:30 am to 12:30 pm in person and in office (No Zoom, in person ONLY)

Monday 12:30 pm to 1:30 pm (No Zoom, in person ONLY)

Email:
Special Request (please help me out – thanks in advance):
My email inbox receives too many emails. It becomes very difficult to manage (and I may miss your email). When you mail me, please visit Canvas first and use the Canvas email system to email. Please **don't** use my email address (nchen@fullerton.edu) to email me. This allows me to clearly see emails from my classes (typically, I have more than 150 students per semester). I understand that this will take you one or two minutes to open Canvas first. I appreciate your help and your kind consideration (your one minute may end up saving me 150 minutes).
Q. I ended up using your email address (instead of the Canvas email system) to email you. Will you reply to my email?
A. This will be just a best-effort, no guarantee (my email box just has too many spam; even if I picked your email without missing it, I may not have any idea about who you are (out of my five classes and 150+ students). Please help me out using the Canvas email system!

Prerequisites: CPSC 362 or Computer Science graduate standing

Course Description:

Edge AI principles and applications in real-world scenarios. Explore various open-source software frameworks used for developing Edge AI applications. Explore key design patterns for edge AI apps. Evaluate hardware choices to optimize performance and power consumption for edge devices.

Course Learning Objective:

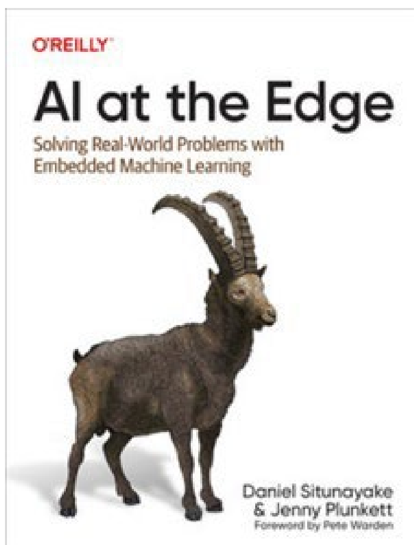
Be able to grow technical competence in Edge AI
Be able to comprehend critical theoretical-level concepts in the field of Edge AI
Be able to develop hands-on skills in implementations and best practices in Edge AI.

Performance Indicators

This course covers the following departmental performance indicators: CODE, COOP, DESC, FB, FDBK, PROC, RESPEC, SPEAK, TEST, WRITE.

Textbooks:

AI at the Edge: Solving Real-World Problems with Embedded Machine Learning
Situnayake, Daniel; Plunkett, Jenny
ISBN 10: 1098120205 / ISBN 13: 9781098120207
Published by O'Reilly Media, 2023



Reading Materials (provided free of charge):

Practical Edge AI – A Logical Questions and Answers Approach
A class note by Ning Chen (In Progress)

Reference Resources (All freely available on the Internet or ACM Digital Library, IEEE Xplore – CSUF students can access freely):

Edge AI Software And Development Tools (https://github.com/TexasInstruments/edgeai)

The Significance of Edge AI towards Real-time and Intelligent Enterprises
(2023 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE) available from IEEE Xplore)

Model training & associated tools (https://github.com/TexasInstruments/edgeai-modelzoo)

Model optimization tools (https://github.com/TexasInstruments/edgeai-torchvision#guidelines-for-model-training--quantization)

Training repositories for various tasks (https://github.com/TexasInstruments/edgeai-mmdetection)

Inference (and compilation) Tools (https://github.com/TexasInstruments/edgeai-tidl-tools)

How to flash OS images to SD cards (https://etcher.balena.io/)

https://git.ti.com/cgit/edgeai/edge_ai_apps

Custom model - compile, benchmark and generate artifacts for deployment on SDK (https://github.com/TexasInstruments/edgeai)

A Gentle Introduction to PyTorch for Beginners (2023) https://www.dataquest.io/blog/pytorch-for-beginners/

Open Compiler for AI Frameworks (NNVM): https://github.com/dmlc/nnvm
TVM (Tensor Virtual Machine): https://github.com/dmlc/tvm

Neural Compute Software Development Kit (provided by the course instructor at no charge per Imagination Technologies' license agreement)

Sample code example 1: Wildlife Monitoring https://github.com/ai-at-the-edge/use-case-wildlife-monitoring

Sample code example 2: Food Quality Assurance https://github.com/ai-at-the-edge/use-case-food-quality-assurance

Sample code example 3: Consumer Products https://github.com/ai-at-the-edge/use-case-consumer-products

Classify any Object using pre-trained CNN Model https://towardsdatascience.com/classify-any-object-using-pre-trained-cnn-model-77437d61e05f

Orange Pi 5 Plus user manual?
https://drive.google.com/drive/folders/1Ov3mZqnMOf_8wpNt9rDxoGIR1ray2iiy

TI (Texas Instruments) Resources:
https://dev.ti.com/edgeaistudio/

ASR (Automatic Speech Recognition)
https://github.com/Uberi/speech_recognition

TinyLlama
https://github.com/airockchip/rknn-llm

https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0

Dataset (Roboflow) and YoloV7

https://blog.roboflow.com/yolov7-custom-dataset-training-tutorial/

YoloV8

https://github.com/laitathei/YOLOv8-ONNX-RKNN-HORIZON-TensorRT-Detection

RKNN Toolkit
https://github.com/airockchip/rknn-toolkit2

https://github.com/airockchip/rknn_model_zoo

Tentative Schedule

(Note: Since the AI field is still evolving, the schedule below is HIGHLY tentative.)

Week (starting day)	Reading materials and Lecture
1 (8/24/2026)	Chapter 1 A Brief Introduction to Edge AI Course orientation and overview Module – Edge AI Course Orientation.ppt Group Project Logistic
2 (8/31/2026)	Module Edge AI Introduction.ppt
3 (9/7/2026)	Chapter 2 Edge AI in the Real World Module – Edge AI in the real world.ppt Group Project Preview Software and Tool Chain Issues

4 (9/14/2026)	Chapter 3 The Hardware of Edge AI Module – Edge AI Hardware.ppt
5 (9/21/2026)	Chapter 4 Algorithms for Edge AI Module – Algorithms for Edge AI.ppt
6 (9/28/2026)	Chapter 5 Tools and Expertise Module – Tools and Expertise.ppt
7 (10/5/2026)	Module – Edge AI Development Workflow (Texas Instruments).ppt Optional readings: Chapter 7 How to Build a Dataset Chapter 8 Designing Edge AI Applications
8 (10/12/2026)	Module – Handwritten Digit Recognition (MNIST classification with CNN on Colab).ppt Optional reading: Chapter 9 Developing Edge AI Applications
9 (10/19/2026)	Module – CIFAR-10 Image Classification on Colab.mp4 Optional reading: Chapter 10 Evaluating, Deploying, and Supporting Edge AI Applications
10 (10/26/2026)	Module – YOLOv7 Overview and Playgound.ppt
11 (11/2/2026)	Module – ARM RK3588 single board computer.ppt
12 (11/9/2026)	Module – Inexpensive Edge AI (RK3588 and YOLOv8 example).ppt
13 (11/16/2026)	Module – Automatic Speech Recognition (ASR) OpenAI Whisper.ppt
14 (11/23/2026)	Fall Recess (No Classes)
15 (11/30/2026)	Module – LLM on inexpensive edge devices.ppt
16 (12/7/2026)	Module – Exploring source code of an LLM (minGPT).ppt
17 (12/14/2026)	Final Exam Week (Hands-On Take Home Final Exam)

Grading Policy:

Grading is based on total points earned.

Discussions (on Canvas): 20%

Attendance (Mandatory*): 20%

Group Project: 10%

Final Exam (online, take-home, hands-on, open-book): 50%

Grades will be assigned at the highest category achieved as follows:

- A+: >=97% of the total points earned
- A: >=93% of the total points earned
- A-: >=90% of the total points earned
- B+: >=87% of the total points earned
- B: >=83% of the total points earned
- B-: >=80% of the total points earned
- C+: >=77% of the total points earned
- C: >=73% of the total points earned
- C-: >=70% of the total points earned
- D: >=60% of the total points earned
- F: < 60% of the total points earned

No late assignments will be accepted. There are no make-ups for missed exams. No extra credit will be given.

***Attendance (Mandatory):**

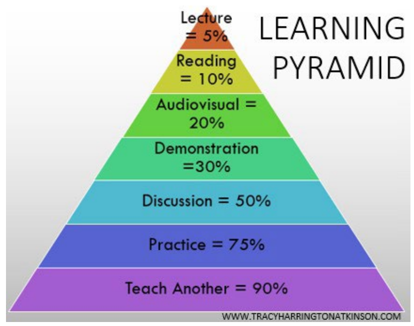
70-100% attendance: 10% (this gives you some buffer for sickness, lateness and emergency; NO need to email me and get permission)

Below 70% attendance: 0% (again, attendance is MANDATORY; if you missed so many attendances, there are NO excuses)

Warning: Roll call occurs at the start of class—no exceptions. Please use your 30% buffer for late arrivals. Re-taking attendance wastes valuable instructional time.

All lectures (mp4 and slides) are available on Canvas. The class time will be reserved for small group discussions. Attendance is mandatory.

(Why small group discussions? Ans: Learning Pyramid)



Per the university policy, twenty percent or less of class meeting time can be taught in an asynchronous online fashion if needed. If it occurs, an announcement email will be issued in advance.

Please note that all assignments are tentative and subject to changes.

Discussion Assignments (© 2026 Ning Chen, All Rights Reserved):

Discussion Assignment - Acknowledgment Statement

You must post (just type or copy-and-paste) the following:

I have reviewed the following:

1. CPSC 476 syllabus
2. FAQs related to discussion/essay/project assignments
3. https://www.plagiarism.org/understanding-plagiarism/ (and I understand the severe consequences resulting from plagiarism and missing deadlines.)

(Type your name here)

Discussion Edge AI (Embedded System/AIoT/IoT/Firmware) Jobs and Their Required Skill Sets

There are many jobs in the field of Edge AI (Embedded System/AIoT/IoT/Firmware). Students who consider seeking those jobs may want to research the job wanted advertisements and analyze the required key skills and qualifications. You are required to post your findings and insights on this topic.

FAQs:

Q. Edge AI is a new buzzword. I couldn't find too many jobs with that title. What should I do?

A. The old buzzwords are embedded system development, IoT (Internet of Things), AIoT (Artificial IoT), Firmware Development. You may also want to use those keywords for your search.

Q. How do I start this discussion?

A. You should search some major job sites (e.g., Monster.com and indeed.com) with some keywords. For example, you may want to use some major keywords such as AI, Edge AI, AIoT, embedded System Development, Firmware Development …

Q. I am not sure there are lots of AI jobs so far. May I focus on some AI startups to see what jobs they offer?

A. Yes (actually AI startups are more interesting).

Q. What should I write?

A. You can write anything as long as it relates to the job descriptions you found and their required skill sets.

Q. Should I also analyze the pay scale of those jobs?

A. Yes, if you can find enough information on the pay scale of the software testing industry.

Q. How do you grade my discussion?

A. The grading of all the discussion assignments is pass or fail (pass – 1 point, fail – 0 point).

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Q. I notice that you want us to have a section entitled "My Closing Thoughts." May I just use the title "Closing Thoughts" without the term "My"?

A. No. This class encourages everyone to develop "independent thinking." Your honest thinking (may not be politically correct or mainstream) is the most valuable things you can keep. Please don't present a summary of other people's thoughts.

Discussion What are the major differences between Edge AI and Cloud AI?

Understanding of those two terms is the starting point of our endeavor. Please research this topic.

FAQs:

Q. How do I write my research?

A. Simply search the Internet and read, read and read.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 2 (Edge AI in the Real World) of the textbook

FAQs:

Q. What should I post for this discussion?

A. You need to read Chapter 2 and summarize your readings.

Q. Do I need to also search the Internet to find additional information?

A. Yes (since you need to provide at least two references to meet the grading rubric).

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort	At least 1000 words	Does not meet the minimum

(You can write whatever you see fit)		
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 3 (The Hardware of Edge AI) of the textbook

FAQs:

Q. What should we do for this discussion?

A. You need to read Chapter 3. Please summarize your readings.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 4 (Algorithm for Edge AI) of the textbook

FAQs:

Q. What should we do for this discussion?

A. You need to read Chapter 4. Please summarize your readings.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize the Orange Pi 5 Plus user manual

FAQs:

Q. What should we do for this discussion?

A. You need to read the Orange Pi 5 Plus User Manual. Please summarize your readings.

Q. Where do I download the Orange Pi 5 Plus user manual?

A. https://drive.google.com/drive/folders/1Ov3mZqnMOf_8wpNt9rDxoGIR1ray2iiy

(If the above link failed, please search yourself – it is very easy to locate it.)

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion ASR (Automatic Speech Recognition)

FAQs:

Q. What should we do for this discussion?

A. You need to study/research the following:

the lecture (ppt and mp4),

whisperMLKDemo.jpynb,

LibriSpeech.ipynb

https://github.com/Uberi/speech_recognition

Q. Where do I find the above?

A. Please review the lecture (the lecture mentions all the above)

(If the github link failed, please search yourself – it is very easy to locate it.)

Q. What should I cover in my discussion?

A. See next question (basically, write something about your study/research on the subject and have "My Closing Thoughts."

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Trying out an LLM

FAQs:

Q. What should we do for this discussion?

A. You need to study/research the following:

the lecture (ppt and mp4),

Paper: Attention Is All You Need

Paper: TinyLlama

https://github.com/airockchip/rknn-llm

https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0

Q. I need to know more about the above topics. Where do I start?

A. Please review the lecture (the lecture mentions all the above)

(If the github link failed, please search yourself – it is very easy to locate it.)

Q. What should I cover in my discussion?

A. See next question (basically, write something about your study/research on the subject and have "My Closing Thoughts."

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 points)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task

Q. What do you mean by Group Project?

A. Members in the assigned group work together to achieve the task. Members can share everything freely.

Q. How do I submit my group project work?

A. EVERY group member needs to submit a copy of the project result.

Q. I don't quite get it. Why do you want so many identical copies (since every group member has the same copy)?

A. People who don't participate in the group effort may not have a copy (we don't want to have free riders).

Group Project Task - Edge AI resources offered by TI (Texas Instruments)

FAQs:

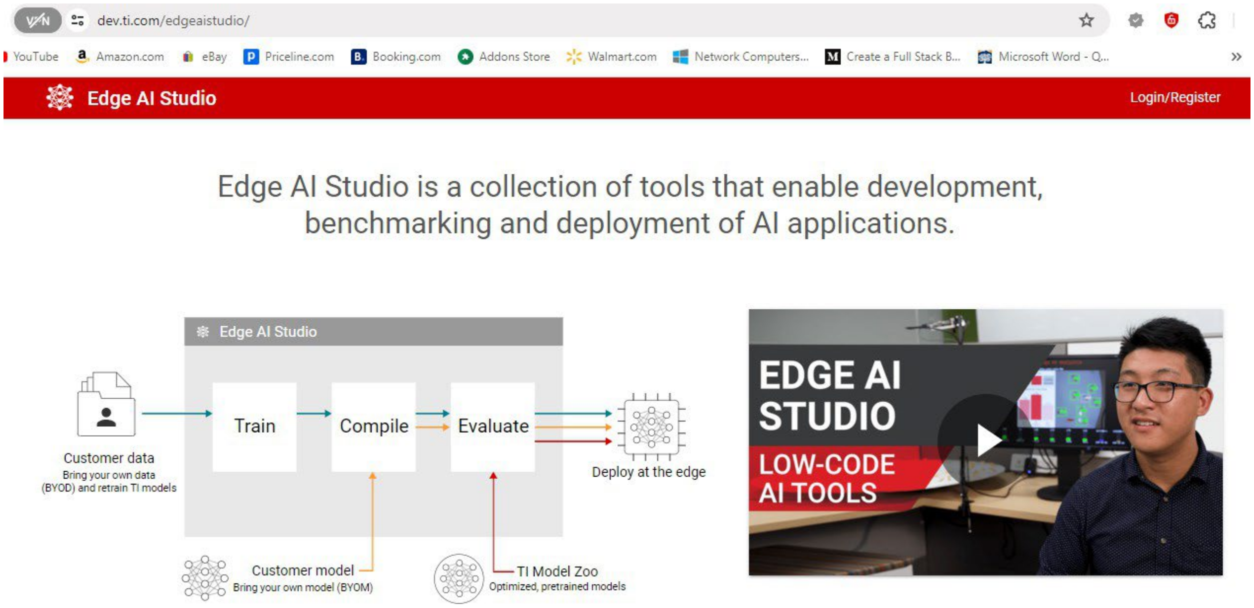
Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by TI.

Q. What does our group need to do?

A. Your group needs to do the following:

Step 1: Visit https://dev.ti.com/edgeaistudio/

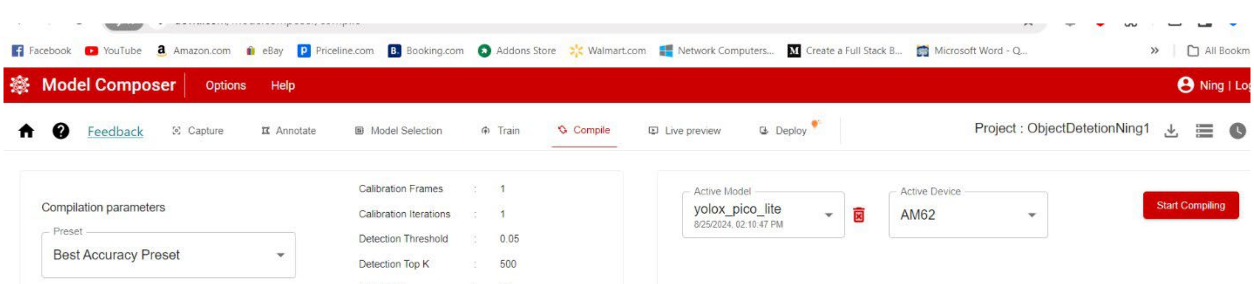


Step 2: Register a free TI account (note that your group may need to have a non-csuf, business.com email account – please overcome that huddle). Q. Our group can't register. What should we do? A. Please see me and I will assign another task for your group.

Step 3: Review two tutorials (Image Segmentation and Object Detection at https://dev.ti.com/edgeaistudio/)

Step 4: Play the Model Composer to reproduce two tutorials (one for Object detection and one for Image Classification).

Please note that since we don't have TI hardware board, your tutorials only needs to go through



Step 5: What to submit for this group assignment

Your group needs to submit two tutorials (two mp4 files) – one for Object Detection and the other for Image Classification. Q. What should we do in our tutorials? A. Just imitate TI's tutorials. The file names should be yourGroupNameOD.mp4 and yourGroupNameIC.mp4.

Group Project Task - Edge AI resources offered by YOLOv7 + roboflow + Google Colab

FAQs:

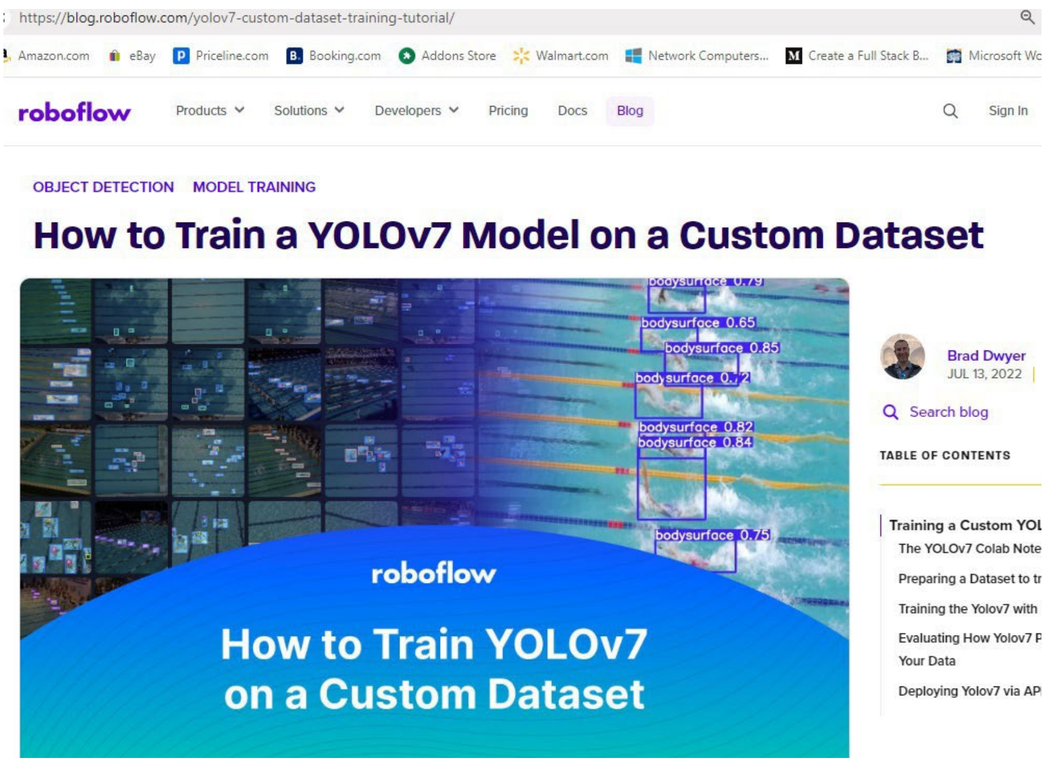
Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by YOLOv7 + roboflow + Google Colab.

Q. What does our group need to do?

A. Your group needs to do the following:

Step 1: Visit https://blog.roboflow.com/yolov7-custom-dataset-training-tutorial/



Step 2: Register a free Google account (your Google account will allow you to run google colab).

Step 3: Duplicate (or imitate) this tutorial and produce your group's own Jupyter Notebook program.

Q. Our work is almost identical to the original tutorial. Is that acceptable?

A. Yes. We try to imitate and learn from other people. Please note that your group may encounter minor issues (mostly relate to data). Please fix them yourself (not too hard).

Step 4: Submit a video (mp4) demo of your group's Jupyter Notebook. The file name should be yourGroupNmaeYOLOv7.mp4. (yes, this is the only file your group needs to submit.)

Group Project Task - Edge AI resources offered by laitatheiYOLOv8-ONNX-RKNN-HORIZON-TensorRT-Detection

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by YOLOv8 model + rknn toolkit2 + python + rk3588 + orangpi 5 plus + lots of others.

Q. Wow! We are not sure we can achieve all those in a very short given time. What should we do?

A. Try to set up a playground (you need to set up rknn toolkit2) and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit https://github.com/laitathei/YOLOv8-ONNX-RKNN-HORIZON-TensorRT-Detection

https://github.com/airockchip/rknn-toolkit2

https://github.com/airockchip/rknn_model_zoo

Step 2: Set up your Linux (Ubuntu) OS (under Windows OS is possible, but you are on your own)

Step 3: Duplicate and run this open-source project (by laitathei, the first link provided above) on your own computer (laptop/desktop)

(Please review the lecture; you can install rknn toolkit2 and do the model conversion from onnx to rknn; your mp4 as an evidence should show that you have done the conversion; you can also use the onnx model to do object detection on your laptop (x86 computer) - please see the lecture, try it anyway; if failed, no worries.)

Step 4: Run the inference task on RK3588 using the rknn model (with the instructor's support in class – no need to do it yourself)

Step 5: Each group needs to submit a presentation that includes three files (1. Report in PDF, 2. PPT file, 3. Mp4 file)

Q. What should we write in PDF?

A. PDF - must have 2000 words; some general review of the yolov8, rknn tollkit2, rk3588, My Closing Thoughts (Actually, My Group's Closing Thoughts) and at least two references.

Q. What should we write in PPT?

A. A condensed version of you PDF.

Q. What should we present in mp4?

A. About 20 minutes of video that goes through you PPT and some screen video demo (just like the class lecture).

Q. What should we include in our screen video demo?

A. You need to at least show that you can do model conversion (from onnx to rknn). This proves that your group went through the rknn toolchain installation and usage. I encourage you to do object detection of a given photo on your laptop directly in the toolchain environment (my lecture mp4 has demonstrated it; you may need to download rnkk model zoo). Please try it (if it failed, that's fine - no deduction of grades).

Group Project Task - Handwritten Digit Recognition (MNIST classification with CNN on Colab)

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study MNIST handwritten digits with CNN on Colab.

Q. Wow! That's exciting but very demanding (we know nothing about NMIST + CNN + Pytorch + Colab). Where do we start?

A. Try to set up a playground and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit and Read

URL 1:

https://www.youtube.com/watch?v=gBw0u_5u0qU

PyTorch Course (2022), Part 4: Image Classification (MNIST)

URL 2

https://github.com/lukepolson/youtube_channel/blob/main/Python%20Tutorial%20Series/pytorch4.ipynb

URL 3

https://github.com/github-bowen/BUAA-ML-MNIST-Classification/blob/main/MNIST-classification-with-CNN/train-on-google-colab.ipynb

Q. Can we find other URLs and learn more?

A. Yes. Of course.

Step 2: Imitate

In Step 2 you need to run the train-on-google-colab.ipynb (jupyter notebook) on your own google colab account.

Q. Is hard to learn colab?

A. No (but no way to avoid dumping hours – it is a good investment).

Q. Will train-on-google-colab.ipynb run on my google colab account with no issues?

A. No. You do need to resolve some issues yourself.

Q. Do you have a read-to-run train-on-google-colab.ipynb? If yes, can you give it to us?

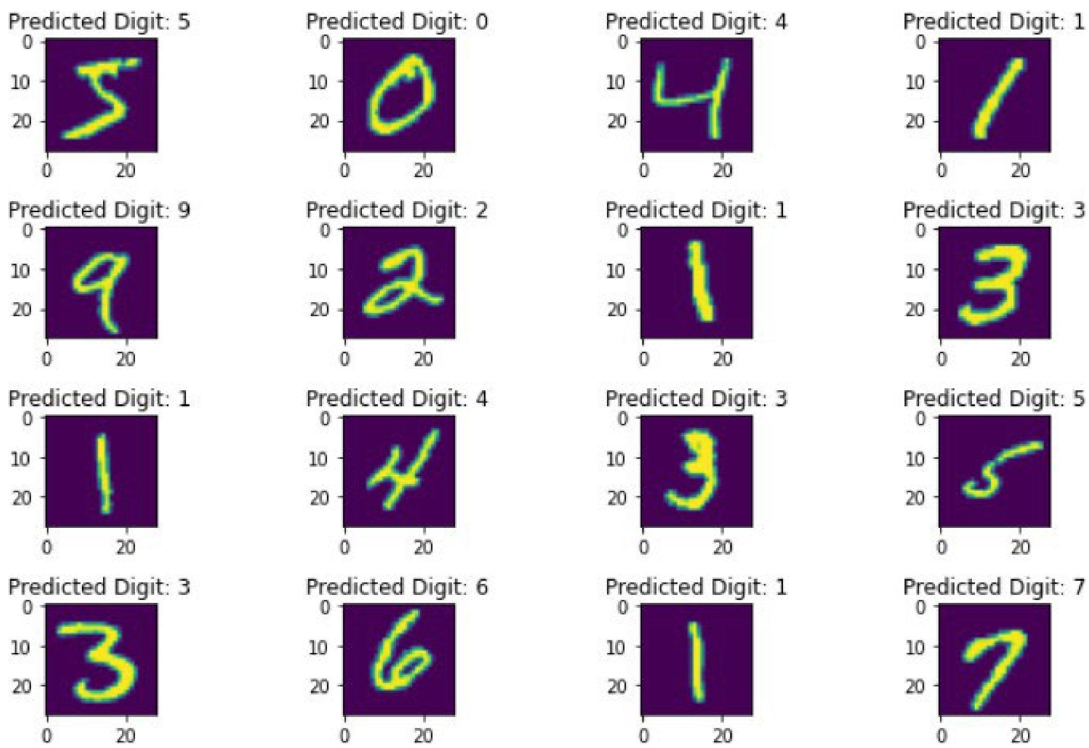
A. Yes. I do have one (after about 10 hours of fixing). Sorry, I don't think giving it to you helps you in any way. Struggling through it is probably the most effective/fastest way of learning.

Step 3: Innovate (just one baby step)

In Step 3 you need to improve the train-on-google-colab.ipynb (you can change its name as you see fit/like) a bit on your own google colab account.

Q. What kind of improvements?

A. URL 2 gives some nice plots (for example, see below). You need to add some of those plots to your jupyter notebook (as you see fit/like – it is your task; you are the boss).



Step 4: Innovate (another small baby step)

In Step 4, you need to improve your version of the jupyter notebook a small step further.

Q. What is another small baby step?

A. You need to save the pytorch model (you trained before) in your local drive (your laptop/computer) first. Add a new section in your notebook to read the saved model (pre-trained model). Run several test cases using the pre-trained model.

Q. What is the purpose of Step 4?

A. After finishing Step 4, you can use other people's pre-trained models to make interesting inferences.

Step 5: Deliverables

In Step 5, you (your group) need to submit the following:

Your jupyter notebook, mp4 (video) that demonstrates the running of your jupyter notebook.

Group Project Task - Training a Classifier (CIFAR10 dataset with CNN on Colab)

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the tutorial of CIFAR10 dataset with CNN on Colab.

Q. Wow! That's exciting but very demanding (we know nothing about CIFAR10 + CNN + Pytorch + Colab). Where do we start?

A. Try to set up a playground and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit and Read

URL 1: https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

(You are welcome to search/read other URLs you can find.)

Step 2: Download the Jupyter Notebook file of this tutorial to your local computer. You can use the download link: https://pytorch.org/tutorials/_downloads/4e865243430a47a00d551ca0579a6f6c/cifar10_tutorial.ipynb (If the link fails, please use URL 1 – you can find the download link on that page.)

Q. Can I use my Google Drive to save the file (instead of using my own local computer)?

A. It is not recommended. For this assignment, please save the Jupyter Notebook file on your own local computer (since you can study/modify it easily).

Step 3: Run the Jupyter Notebook file on your colab account

Q. How do I do that?

A. There are many different ways. An easier way probably is to log in to your gmail account first and then use your browser to search for colab (and click the link, then you will be on colab).

Q. How do I upload my local Jupyter Notebook file to colab?

A. File -> Upload notebook

Q. How do I run my notebook file?

A. Please just google a bit.

Step 4: Run the notebook (it is bugfree; very easy to run – just enjoy it)

Step 5: Study/research the class net (shown below)

```
import torch.nn as nn
import torch.nn.functional as F


class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 6, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16 * 5 * 5, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = torch.flatten(x, 1) # flatten all dimensions except batch
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = self.fc3(x)
        return x
```

Step 4: Deliverable

Each group needs to submit a research paper (in pdf) that teaches readers about this class.

Q. What should our group write?

A. There are many things your readers may not understand. For example, they may not have any ideas on Cov2d, MaxPool2d, Liner (fc1, fc2, fc3, why there are three of them), pool, relu, flatten.

Q. Can we include figures and photos?

A. Yes.

Q. Any minimum requirements?

A. Yes. Your report should have at least 5 pages long – title page won't count (more pages are welcome).

Q. Any required format/style?

A. No. You can write whatever you see fit.

Q. Do we need to say anything about the rest (running the notebook, image classification, training, accuracy, etc)?

A. No. In this assignment, we only focus on the model architecture (structure) itself.

Alternative procedures for submitting work: In case of technical difficulties with the Canvas system or other online resources, your instructor will communicate with students directly through their CSUF email. The instructor will provide directions on alternative methods for submitting work and/or may extend submission deadlines. If email is not available, students may contact the department office for guidance.

Authentication of student work: All assignments submitted for a grade in this class must be your individual work unless otherwise indicated.

Student Support and Information: All course materials are posted on Canvas. All student emails are managed on Canvas and responded to within two business days.

Technical Support:
If you encounter any technical difficulties, contact the instructor immediately to document the problem. Then, contact: student IT help desk, email, phone (657) 278-8888, walk-in student genius center, online chat - log into portal; click "Online IT Help"; click "Live Chat."
For issues with Canvas: Canvas Support Hotline (855) 302-7528, student support chat
Course-related issues – please contact the course instructor (via the Canvas email system)

Technical Competencies:
If you met the prerequisite requirements (CPSC 362 or Computer Science graduate standing), no other technical competencies are required.

Additional Information:

Student Civility:

Please see UPS 100.006 (http://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20100/UPS%20100.006.pdf)

Students with Special Needs:

Please visit Disability Support Services (https://www.fullerton.edu/dss/)

Academic Dishonesty Policy:

Please see UPS 300.021 (https://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20300/UPS%20300.021.pdf)

Violations of UPS300.021 will result in an F grade.

For your convenience, parts of the UPS 300.021 are shown below:

"Academic dishonesty includes such things as cheating, inventing false information or citation, plagiarism, and helping someone to commit an act of academic dishonesty. It usually involves an attempt by a student to show possession of a level of knowledge or skill, which he/she, in fact, does not possess.
Cheating is defined as the act of obtaining or attempting to obtain credit for work by the use of any dishonest, deceptive, fraudulent, or unauthorized means. Examples of cheating include, but are not limited to, the following: using notes or aids or help of other students on tests and examinations in ways other than those expressly permitted by the instructor, plagiarism as defined below, tampering with grading procedures, and collaborating with other on any assignment where such collaboration is expressly forbidden by an instructor. Violation of this prohibition of collaboration shall be deemed on offense for the person or persons collaborating on the work, in addition to the person submitting the work.
Plagiarism is defined as the act of taking the work of another and offering it as one's own without giving credit to that source. When sources are used in a paper, acknowledgment of the original author or source must be made through appropriate references and, if directly quoted, quotation marks or indentations must be used"

Emergency Preparedness:
Please visit the Emergency Preparedness website (https://police.fullerton.edu/programs/prepare/).

CSUF Police Department
Dispatch:
(657) 278-2515
Hours: 24/7

Withdrawal of the course:

Please see Withdrawal FAQs (http://records.fullerton.edu/services/withdrawal.php).
Withdrawal deadline: Please see the above Withdrawal FAQs.

Student Resources Website:
It is the student's responsibility to read and understand the required and important student information for course syllabi. Included is information about:

- University learning goals
- General Education learning objectives

- Netiquette/appropriate online behavior
- Students' rights to accommodations
- Campus student support resources
- Academic integrity
- Emergency preparedness/what to do
- Library services
- Student IT services and competencies
- Software privacy and accessibility
- Accessibility statement
- Diversity statement
- Land acknowledgement
- Final exam schedule
- Semester calendar